



University of Engineering and Technology, Mardan

Week 1 Task Report

Titanic Dataset – Python EDA

Career Launchpad Phase II

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Week 1: Data Analysis & Visualization — Titanic Dataset (Python)

Introduction

This task was completed as part of Career Launchpad Phase 2, Week 1, under the Data Analysis & Visualization domain. The objective was to perform a complete workflow of data import, cleaning, preprocessing, exploratory analysis, and visualization on the Titanic dataset using Python in Google Colab, and to present the findings with supporting charts.

Data Cleaning & Preprocessing

The dataset was imported into a Pandas DataFrame and prepared for analysis:

- **Dataset Info & Structure:** Checked column data types, dataset shape, and overall structure.
- **Missing Values:** Identified null values, particularly in the Age and Cabin columns.
- **Duplicates:** Checked and removed duplicate rows to preserve data integrity.
- **Missing Value Treatment:** Filled missing numeric values (such as Age) using column means.
- **Index Reset:** Reset the DataFrame index after cleaning for consistency.

Exploratory Data Analysis (EDA)

A detailed exploratory analysis was carried out to understand passenger demographics and survival patterns:

- **Survival Count:** 549 passengers did not survive while 342 survived, showing that the majority of passengers did not make it. (Figure 1)
- **Gender Distribution:** The dataset contained 577 male and 314 female passengers. (Figure 2)
- **Passenger Class Distribution:** Class 3 had the highest number of passengers (491), followed by Class 1 (216) and Class 2 (184). (Figure 3)
- **Survival by Gender:** Female passengers had a notably higher survival rate than male passengers, with most male passengers not surviving. (Figure 4)
- **Age Distribution:** The age distribution is right-skewed, with the largest concentration of passengers between 20–40 years. (Figure 5)
- **Fare Distribution:** Most fares were concentrated below 50, with a long right tail of a few high-value fares. (Figure 6)
- **Age vs Survival:** A boxplot comparison showed similar median ages for survivors and non-survivors, though non-survivors included more older outliers. (Figure 7)
- **Multivariate Relationships:** A pairplot of Age, Fare, and Pclass against Survived highlighted how lower-class, higher-fare and younger passengers related to survival outcomes. (Figures 8–9)

Data Visualization

The following charts were generated in Python (Seaborn/Matplotlib) to support the analysis:

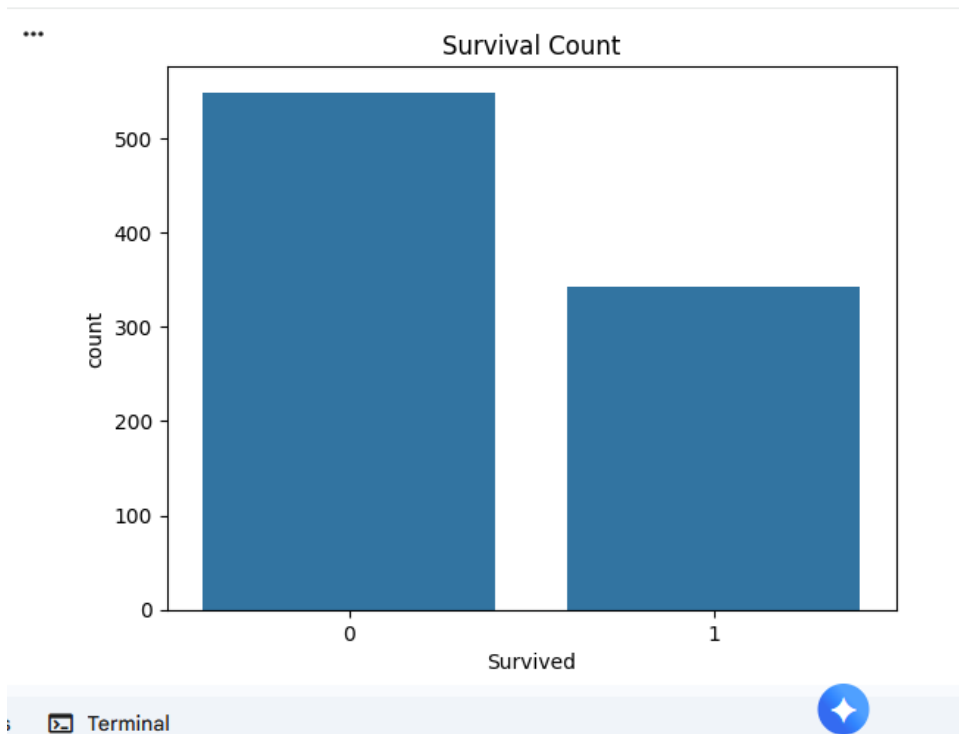


Figure 1: Survival Count

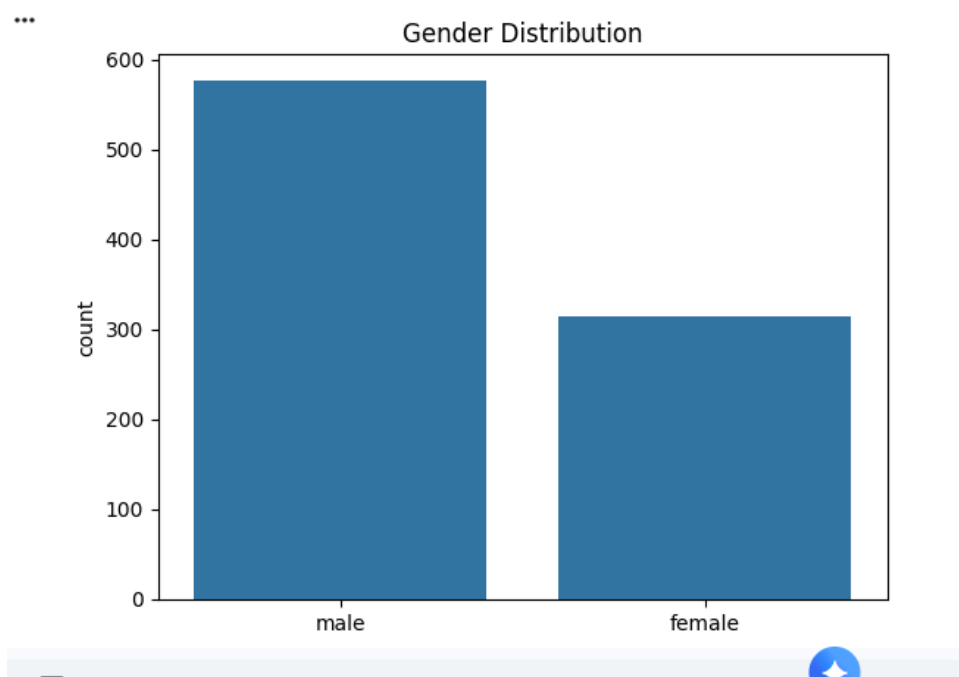


Figure 2: Gender Distribution

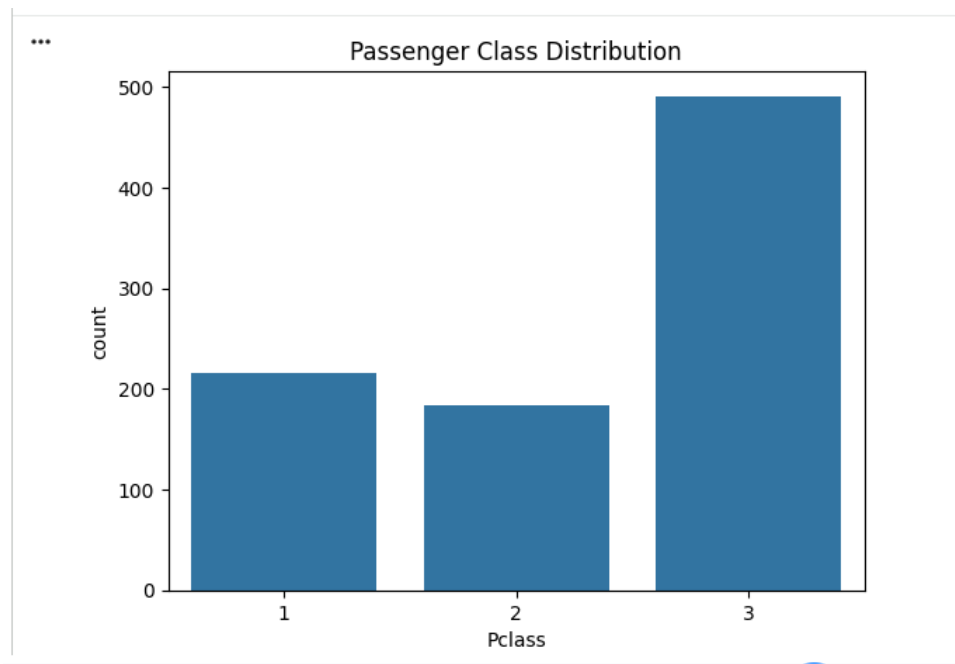


Figure 3: Passenger Class Distribution

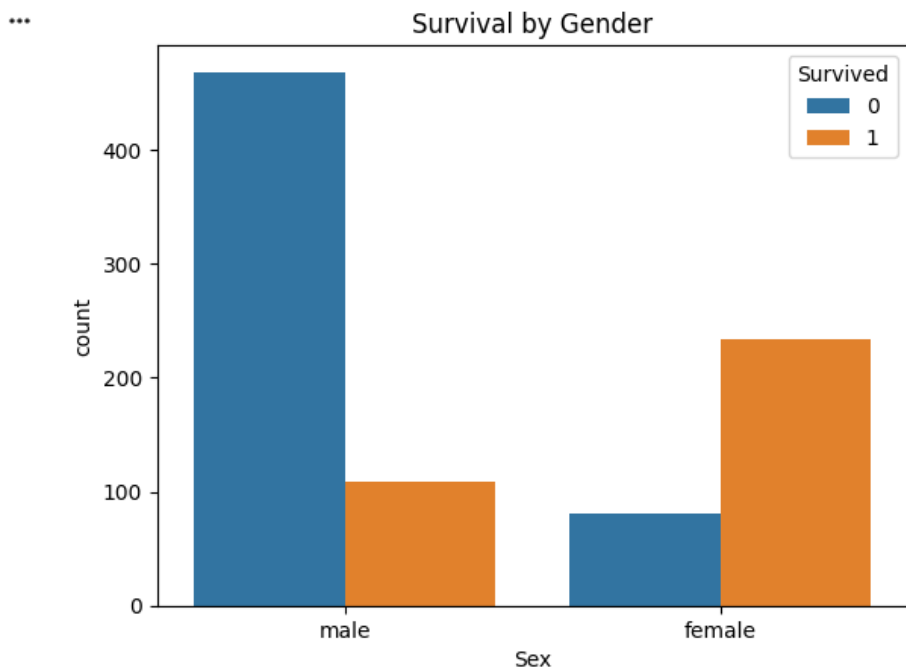


Figure 4: Survival by Gender

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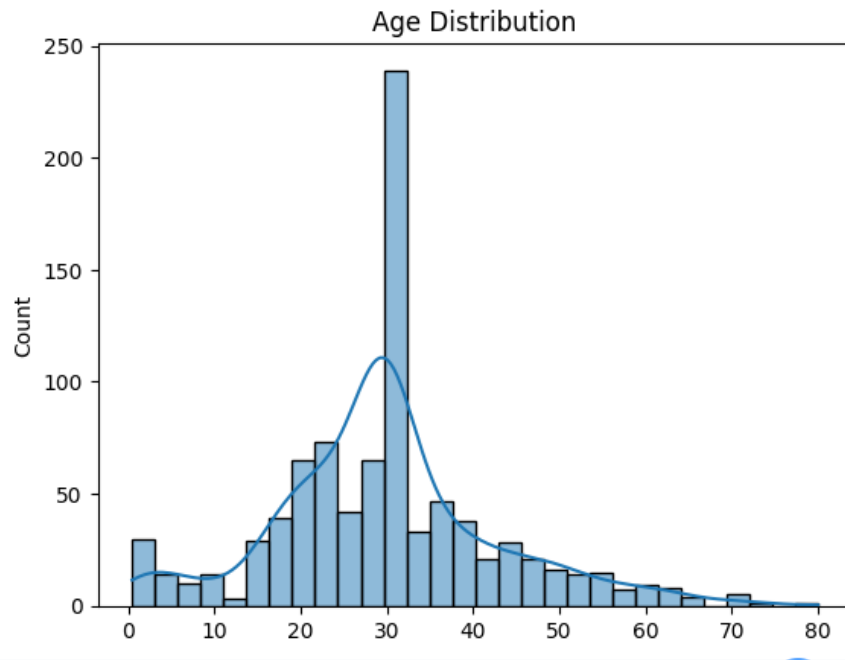


Figure 5: Age Distribution

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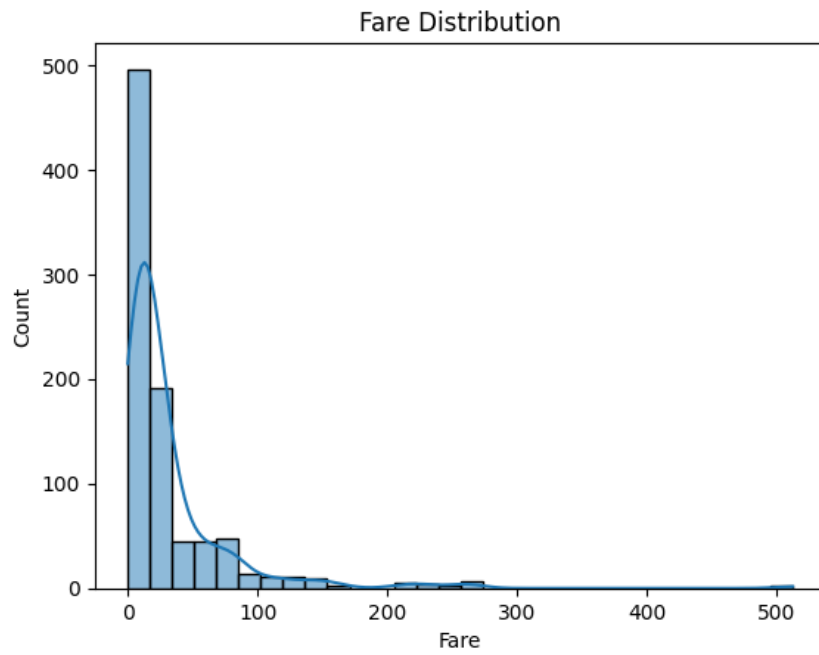


Figure 6: Fare Distribution

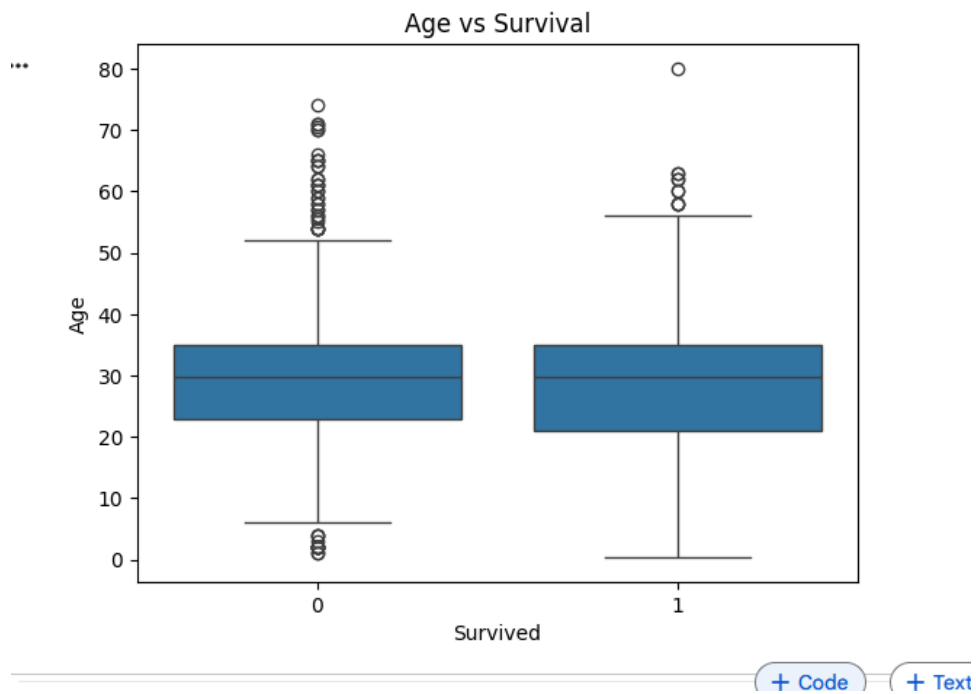


Figure 7: Age vs Survival

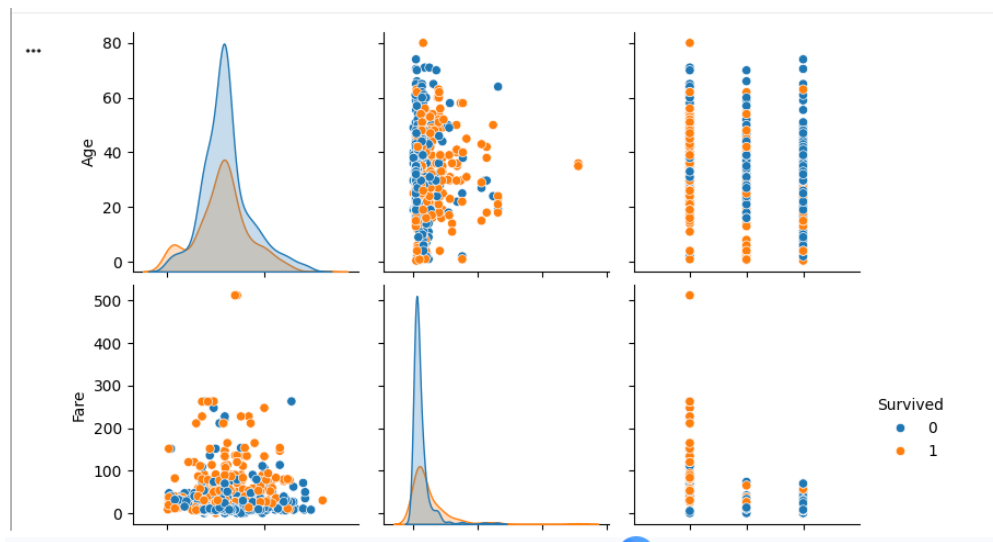


Figure 8: Pairwise Relationships (Age, Fare, Pclass vs Survived)

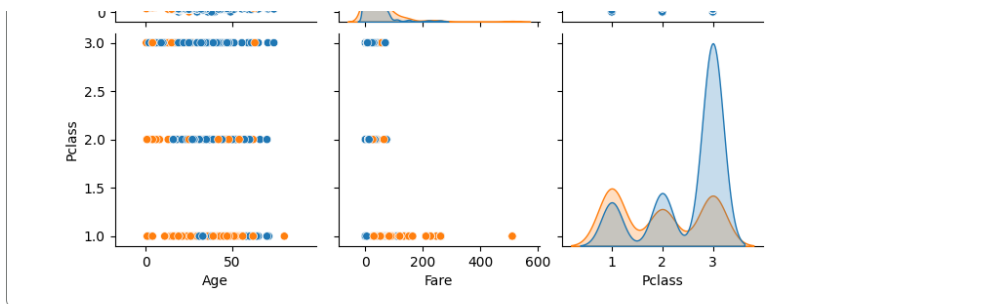


Figure 9: Pairwise Relationships (continued)

Key Insights

- Survival rate was significantly higher among female passengers than male passengers.
- Passenger class strongly influenced survival, with first-class passengers more likely to survive.
- Most passengers were young to middle-aged adults, with fares heavily skewed toward lower values.
- The dataset required only minor cleaning and is well suited for further predictive modeling.

Conclusion

Week 1's analysis of the Titanic dataset in Python confirmed that survival was strongly linked to gender and passenger class. The cleaning, exploratory analysis, and visualization workflow provided a solid foundation for further statistical or predictive work on this dataset.